

Dt : 9/2/2023

***imp**

JSP Programming:(Unit-3)

=>JSP Stands for 'Java Server Page' and which is response from WebApplication.

=>JSP is tag based programming language and which is more easy when compared to Servlet programming.

=>Programs in JSP are saved with (.jsp) as an extention.

=>JSP programs are combination of both HTML code and Java Code.

=>JSP provides the following tags to write JavaCode part of JSP programs:

1.Scripting tags

=>Scriptlet tag

=>Expression tag

=>Declarative tag

2.Directive tags

=>page

=>include

=>>taglib

3.Action tags

=>jsp:include

=>jsp:forward

=>jsp:param

=>jsp:useBean

=>jsp:setProperty

=>jsp:getProperty

1.Scripting tags:

=>Scripting tags are used to write JavaCode part of JSP programs.

=>Scripting tags are categorized into the following:

(a)Scriptlet tag

(b)Expression tag

(c)Declarative tag

(a)Scriptlet tag:

**=>Scriptlet tag is used to write normal JavaCode(Servlet Code)
part of JSP programs**

syntax:

<% ---JavaCode--- %>

(b)Expression tag:

=>Expression tag is used to assign the value to variable or

which is used to display the data to the WebBrowser.

syntax:

<%= expression %>

(c)Declarative tag:

=>Declarative tag is used to declare variables and methods in JSP programs.

syntax:

<%! variables;methods %>

=====

2.Directive tags:

=>The tags which are used to specify the directions in translation process are known as Directive Tags.

The following are the types of Directive tags:

(a)page

(b)include

(c)taglib

(a)page:

=>'page' directive tag specifies the translator to add the related attribute to current JSP page.

syntax:

<%@ page attribute="value" %>

exp:

<%@ page import="java.util.*"%>

List of attributes:

1.import

2.contentType

3.extends

4.info

5.buffer

6.language

7.isELIgnored

8.isThreadSafe

9.autoFlush

10.session

11.pageEncoding

12.errorPage

13.isErrorPage

(b)include:

=>'include' directive tag specifies the file to be included to current JSP page.

syntax:

<%@ include file="file-name"%>

Exp:

<%@ include file="input.html" %>

(c>taglib:

=>'taglib' directive tag specifies to add specified url to current JSP page and which is used part of EL(Expression Lang) and JSTL (JSP Standard Tag Lib).

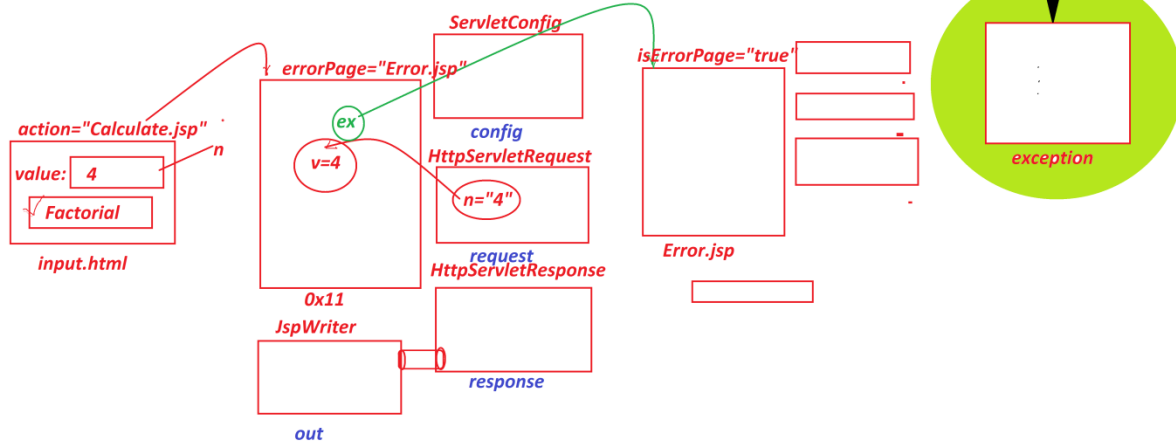
syntax:

`<%@ taglib url="urloftaglib" prefix="prefixoftaglib"%>`

=====

Ex : JSP Application

Layout:



IF ANY USER ENTER FLOTING VALUES(4.5) EXCEPTION WILL GENERATE for that we USED "ISERRORPAGE"

input.html

```
<!DOCTYPE html>
<html>
<head>
<meta charset="ISO-8859-1">
<title>Insert title here</title>
</head>
<body>
  <form action="Calculate.jsp" method="post">
    Enter the Value:<input type="text" name="n"><br>
    <input type="submit" value="Factorial">
  </form>
</body>
</html>
```

```
    </form>
</body>
</html>
```

Calculate.jsp

```
<%@ page language="java"
    contentType="text/html; charset=ISO-8859-1"
    pageEncoding="ISO-8859-1"
    errorPage="Error.jsp"%>
<!DOCTYPE html>
<html>
<head>
<meta charset="ISO-8859-1">
<title>Insert title here</title>
</head>
<body>
<%!
int fact;
int factorial(int n)
{
    fact=1;
    for(int i=n;i>=1;i--)
    {
        fact=fact*i;
    }
    return fact;
}
%>
<%
int v = Integer.parseInt(request.getParameter("n"));
int res = factorial(v);
out.println("Factorial:"+res+"<br>");
%>
<%@include file="input.html"%>
</body>
</html>
```

web.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<web-app>
  <welcome-file-list>
    <welcome-file>input.html</welcome-file>
  </welcome-file-list>
</web-app>
```

Error.jsp

```
<%@ page language="java"
      contentType="text/html; charset=ISO-8859-1"
      pageEncoding="ISO-8859-1"
      isErrorPage="true"%>
<!DOCTYPE html>
<html>
<head>
<meta charset="ISO-8859-1">
<title>Insert title here</title>
</head>
<body>
<%
out.println("Enter only Integer value...<br>");
%>
<%= exception %>
<br>
<%@include file="input.html"%>
</body>
</html>
```

=====

=>The following are the implicit objects of JSP:

application - javax.servlet.ServletContext

config - javax.servlet.ServletConfig

request - *javax.servlet.http.HttpServletRequest*
response - *javax.servlet.http.HttpServletResponse*
out - *javax.servlet.jsp.JspWriter*
session - *javax.servlet.http.HttpSession*
exception - *java.lang.Throwable*
page - *java.lang.Object*
pageContext - *javax.servlet.jsp.PageContext*

=====

3.Action tags:

=>Action Tags are used to include some basic actions like inserting some other page resources ,forwarding the request to another page, creating or locating the JavaBean instances and,setting and retriving the bean properties in JSP pages.

Note:

=>These action tags are used in Execution process at runtime.

=>The following are some Important action tags available in JSP:

- 1.<jsp:include>**
- 2.<jsp:forward>**
- 3.<jsp:param>**
- 4.<jsp:useBean>**

5.<jsp:setProperty>

6.<jsp:getProperty>

=====

1.<jsp:include> :

=>This action tag allows to include a static or dynamic resource such as HTML or JSP specified by an URL to be included in the current JSP while processing request.

=>If the resource is static then its content is included in the JSP page.

=>If the resource is dynamic then its result is included in the JSP page.

syntax:

<jsp:include attributes>

<---Zero or more jsp:param tags--->

</jsp:include>

attributes of include tag:

***imp**

page : Takes a relative URL,which locates the resource to be included in the JSP page.

`<jsp:include page="/Header.html"/>`

`<jsp:include page="<%=mypath%"/>`

flush : Takes true or false, which indicates whether or not the buffer needs to be flushed before including resource.

2.<jsp:forward>:

=> This action tag forwards a JSP request to another resource and which can be either static or dynamic.

=> If the resource is dynamic then we can use a jsp:param tag to pass name and value of the parameter to the resource.

syntax:

`<jsp:forward attributes>`

`<-- Zero or more jsp:param tags-->`

`</jsp:forward>`

Exp:

`<jsp:forward page="/Header.html"/>`

`<jsp:forward page="<%=mypath%"/>`

3.<jsp:param>:

This action tag is used to hold the parameter with value and which is to be forwarded to the next resource.

syntax:

<jsp:param name="paramName" value="paramValue"/>

=====

4.<jsp:useBean>:

=>This tag is used to instantiate a JavaBean, or locate an existing bean instance and assign it to a variable name(id).

syntax:

<jsp:useBean attributes>

<!--optional body content-->

</jsp:useBean>

Attributes of <jsp:useBean> tag:

(a)id

(b)scope

(c)class

(d)beanName

(e)type

(a)id:

=>which represents the variable name assigned to id attribute of <jsp:useBean> tag and which holds the reference of JavaBean instance.

(b)scope:

=>which specifies the scope in which the bean instance has to be created or located.

scope can be the following:

(i)page scope : within the JSP page,until the page sends response.

(ii)request scope : JSP page processing the same request until a JSP sends response.

(iii)session scope - Used with in the Session.

(iv)application scope - Used within entire web application.

***imp**

(c)class :

=>The class attribute takes the qualified class name to create a bean instance.

(d)beanName :

=>The *beanName* attribute takes a qualified class name.

(e)type :

=>The "type" attribute takes a qualified className or interfaceName, which can be the classname given in the class or beanName attribute or its super type.

5.<jsp:setProperty>:

=>This action tag sets the value of a property in a bean, using the bean's setter methods.

Types of attributes:

(a)name

(b)property

(c)value

(d)param

(a)name:

=>The name attribute takes the name of already existing bean as a reference variable to invoke the setter method.

(b)property:

=>which specifies the property name that has to be set,and specifies the setter method that has to be invoked.

(c)value:

=>The value attribute takes the value that has to be set to the specified bean property.

(d)param:

=>which specify the name of the request parameter whose value to be assigned to bean property.

6.<jsp:getProperty>:

=>This action tag gets the value of a property in a bean by using the bean's getter method and writes the value to the current JspWriter.

Types of attributes:

(a)name

(b)property

(a)name:

=>The name attribute takes the reference variable name on which we

want to invoke the getter method.

(b)property:

=>which gets the value of a bean property and invokes the getter method of the bean property.

The following are some rare used Actions tags:

7.< jsp:plugin >:

The <jsp:plugin> action tag provide easy support for including a java applet in the client Web browser, using a built-in or downloaded java plug-in.

Syntax:

<jsp:plugin attributes>

<!--optionally one jsp:params or jsp:fallback tag--

</jsp:plugin>

8. < jsp:fallBack >

The <jsp:fallback> action tag allows us to specify a text message to be displayed if the required plug-in cannot run and this action tag must be used as a child tag with the <jsp:plugin> action tag.

Syntax:

<jsp:fallback>

Test message that has to be displayed if the plugin cannot be started

</jsp:fallback>

9. <jsp:params >

The <jsp:params> action tag sends the parameters that we want to pass to an applet.

Syntax:

<jsp:params>

<!--one or more jsp:param tags-->

</jsp:params>

=====

====